

AFRILANG Stdlib

std/c/graphcolor

Français. Bibliothèque AFRILANG 'std/c/graphcolor' (niveau complex, catégorie complex). Elle expose 7 fonction(s) : greedyColor, chromaticNumber, isValidColoring, colorClasses, conflictCount, welshPowell, sameColorNeighbors.

```
'import "std/c/graphcolor"' · 'use graphcolor'
```

- 'greedyColor(adj list number, n number) → list number' - 'chromaticNumber(adj list number, n number) → number' - 'isValidColoring(adj list number, n number, colors list number) → bool' - 'colorClasses(adj list number, n number) → list number' - 'conflictCount(adj list number, n number, colors list number) → number' - 'welshPowell(ad

English.

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Catégorie / Category	Modules complex (c/)	
Niveau / Tier	complex	June 16, 2026
Fonctions / Functions	7	

Contents

Français

1. Vue d'ensemble

Bibliothèque AFRILANG 'std/c/graphcolor' (niveau complex, catégorie complex). Elle expose 7 fonction(s) : greedyColor, chromaticNumber, isValidColoring, colorClasses, conflictCount, welshPowell, sameColorNeighbors.

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- `greedyColor(adj list number, n number) → list number`
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- `conflictCount(adj list number, n number, colors list number) → number`
- `welshPowell(adj list number, n number) → list number`
- `sameColorNeighbors(adj list number, n number, node number) → number`
```

2. Installation & import

Ajoutez le module à votre programme :

```
import "std/c/graphcolor"
use graphcolor
```

Niveau : complex · Catégorie : Modules complex (c/)
Domaines avancés – graphes, NLP, réseaux complexes.

3. Référence API

- greedyColor(adj list number, n number) → list•
- chromaticNumber(adj list number, n number) → number•
- isValidColoring(adj list number, n number, colors list number) → bool•
- colorClasses(adj list number, n number) → list•
- conflictCount(adj list number, n number, colors list number) → number•
- welshPowell(adj list number, n number) → list•
- sameColorNeighbors(adj list number, n number, node number) → number

4. Exemples d'utilisation

```
import "std/c/graphcolor"
use graphcolor
```

```
create result = greedyColor(/* args */)
say result
```

```
create result = chromaticNumber(/* args */)
say result
```

```
create result = isValidColoring(/* args */)
say result
```

```
create result = colorClasses(/* args */)
say result
```

```
create result = conflictCount(/* args */)
say result
```

```
create result = welshPowell(/* args */)
say result
```

5. Bonnes pratiques

- Préférez ``import "std/c/graphcolor"`` en tête de fichier.
- Lancez ``afriLang check`` avant de livrer.
- Combinez avec les modules `stdlib` de la même catégorie.
- Voir `/stdlib/` pour le source live et les démos playground.
- Signalez les problèmes sur GitHub si une export manque.

6. Annexe — source

module graphcolor

```
function greedyColor(adj list number, n number) returns list number
end
```

```
function chromaticNumber(adj list number, n number) returns number
end
```

```
function isValidColoring(adj list number, n number, colors list number) returns bool
end
```

```
function colorClasses(adj list number, n number) returns list number
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function conflictCount(adj list number, n number, colors list number) returns number
end
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```
function welshPowell(adj list number, n number) returns list number
end
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function sameColorNeighbors(adj list number, n number, node number) returns number
end
end
```

English

1. Overview

AFRILANG library 'std/c/graphcolor' (tier complex, category complex). Exports 7 function(s): greedyColor, chromaticNumber, isValidColoring, colorClasses, conflictCount, welshPowell, sameColorNeighbors.

'import "std/c/graphcolor"' · 'use graphcolor'

```
- `greedyColor(adj list number, n number) → list number`
- `chromaticNumber(adj list number, n number) → number`
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- `welshPowell(adj list number, n number) → list number`
- `sameColorNeighbors(adj list number, n number, node number) → number`
```

2. Installation & import

Add the module to your program:

```
import "std/c/graphcolor"
use graphcolor
```

Tier: complex · Category: Complex modules (c/)
Advanced domains – graphs, NLP, complex networks.

3. API reference

- greedyColor(adj list number, n number) → list•
- chromaticNumber(adj list number, n number) → number•
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- welshPowell(adj list number, n number) → list•
- sameColorNeighbors(adj list number, n number, node number) → number

4. Usage examples

```
import "std/c/graphcolor"
use graphcolor
```

```
create result = greedyColor(/* args */)
say result
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create result = chromaticNumber(/* args */)
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```

```
create result = conflictCount(/* args */)
say result
```

```
create result = welshPowell(/* args */)
say result
```

5. Best practices

- Prefer ``import "std/c/graphcolor"`` at the top of your file.
- Run ``afriLang check`` before shipping.
- Combine with related stdlib modules from the same category.
- See `/stdlib/` for the live source and playground demos.
- Report issues on GitHub if an export is missing or undocumented.

6. Appendix — source

module graphcolor

```
function greedyColor(adj list number, n number) returns list number
end
```

```
function chromaticNumber(adj list number, n number) returns number
end
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function isValidColoring(adj list number, n number, colors list number) returns bool
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function colorClasses(adj list number, n number) returns list number
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function conflictCount(adj list number, n number, colors list number) returns number
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function welshPowell(adj list number, n number) returns list number
end
```

```
function sameColorNeighbors(adj list number, n number, node number) returns number
end
end
```

Référence rapide / Quick reference

- Import : `import "std/c/graphcolor"`
- Vérification : `afriLang check`
- Documentation : <https://afriLang.dev/stdlib/std/c/graphcolor/>

Source (excerpt)

```
module graphcolor

function greedyColor(adj list number, n number) returns list number
end

function chromaticNumber(adj list number, n number) returns number
end

function isValidColoring(adj list number, n number, colors list number) returns bool
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function colorClasses(adj list number, n number) returns list number
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function conflictCount(adj list number, n number, colors list number) returns number
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function welshPowell(adj list number, n number) returns list number
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function sameColorNeighbors(adj list number, n number, node number) returns number
end
end
```